Bauhaus University

Attitude and Orbit Control System (AOCS)

Project

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Date: 1/9/2025

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Submission Date: 27/11/2025

Contents

3D Representation of a spacecraft

- Body Fixed Frame
- Earth Centered Inertial Frame (ECI)
- Local Vertical Local Horizontal Frame
- Southeast Zenit Frame
- Northeast Nadir Frame
- Earth centered Earth Fixed Frame
- Orbit Frame
- Space Craft Coordinate Frame

Rotational Kinematics

The orientation of one frame with respect to another represented in four ways

- Direction Cosine Matrices
- Euler Angles
- Quaternions
- Modified Rodrigues Parameters

Quaternions

Quaternion represents rotation by a rotational angle around a rotational axis.

The quaternion is represented as the sum of scalar and vector

$$\overline{q} = q_0 + q_1 * i + q_2 * j + q_3 * k \tag{1.1}$$

q_0 is the scalar form of a quaternion
$q_1 * i + q_2 * j + q_3 * k$ is the vector form of a quaternion

The quaternion can also be represented in normalized form

$$\overline{q} = \cos\frac{\alpha}{2} + \hat{e} * \sin\frac{\alpha}{2}$$

Quaternion based model has several advantages compared to Euler angle-based model.

- Quaternion based models are uniquely defined because it does not depend on rotational sequence
- A full quaternion Model does not have singular point in a rotational sequence (No gimble lock)
- Use of quaternions has computational efficiency

Equation of Motion of a Space Craft

The cost of using only three components of the quaternion in the model is that, similar to the Euler angle representation, the reduced model has a singular point at $\alpha = \pm\pi$, where α is the rotation angle around the rotation axis. However, this singular point is the farthest point to the point where the linearization is carried out. The stability of the designed closed-loop spacecraft system is guaranteed because the linearized control system is fully controllable.

The dynamics equation of a spacecraft

Let J bet inertia matrix of a Space craft

$$J = \begin{bmatrix} J_{11} & J_{12} & J_{13} \\ J_{21} & J_{22} & J_{23} \\ J_{31} & J_{32} & J_{33} \end{bmatrix} \tag{1.2}$$

The angular velocity vector of a spacecraft with respect to inertia

$$w_I = [w_{I1} \quad w_{I2} \quad w_{I3}]^T \tag{1.3}$$

Angular moment **h** is the product of spacecraft inertia and angular velocity vector

$$h = J * w_I \tag{1.4}$$

The resultant rate of change of angular velocity is Torque T

$$\frac{dh}{dt} = T$$

$$T = \frac{dh}{dt} + w_I \times h \tag{1.5}$$

Following is the resultant equation

$$\frac{dh}{dt} = T - w_I \times (Jw_I) \tag{1.6}$$

The external torque T applied on spacecraft due to some external disturbances like gravitational, aerodynamic, solar radiation, and other environmental torques is denoted as t_d and control torque u

$$t_d = [t_{d1} \quad t_{d2} \quad t_{d3}]^T$$

$$u = [u_1 \quad u_2 \quad u_3]^T$$

The resultant following equation

$$\frac{dh}{dt} = -w_I \times (Jw_I) + t_d + u \tag{1.8}$$

The kinematic equation of a spacecraft

The normalized form of a quaternion is represented below with scaler $q_0 = \cos\frac{\alpha}{2}$ and the vector $q = \hat{e} * \sin\frac{\alpha}{2}$

$$\overline{q} = \cos\frac{\alpha}{2} + \hat{e} * \sin\frac{\alpha}{2}$$

The derivative of quaternion w.r.t to t

$$\frac{d\overline{q}}{dt} = \overline{q}(t) \otimes \left(\frac{1}{2} * \hat{e} * \Omega(t)\right) = \overline{q}(t) \otimes \left(\frac{1}{2} w(t)\right)$$

Ω(t)=limΔt→0 $\frac{\Delta\alpha}{\Delta t}$ is a scalar, and ω(t)= $\hat{e}$(t)Ω(t) is a vector, and(0+ $\frac{1}{2}$ w(t))= $\frac{1}{2}$ (0, w_1, w_2,w_3)is a quaternion.

$$\begin{bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -w_1 & -w_2 & -w_3 \\ w_1 & 0 & w_3 & -w_2 \\ w_2 & -w_3 & 0 & w_1 \\ w_3 & w_2 & -w_1 & 0 \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \tag{1.9}$$

The nonlinear spacecraft kinematics equation of motion can be represented by the quaternion as follows.

$$\begin{cases} \dot{q} = -\dfrac{1}{2} w \times q + \dfrac{1}{2}\, q_0 w \\ \dot{q} = -\dfrac{1}{2} w^T \times q \end{cases}$$

The magnitude of a quaternion vector $q_0{}^2 + q_1{}^2 + q_2{}^2 + q_3{}^2 = 1$

$$q_0 = \sqrt{1 - q_1{}^2 - q_2{}^2 - q_3{}^2}$$

To obtain a controllable quaternion model, only vector component of the quaternion is used in the model. The cost of using only three components of the quaternion in the model is that, similar to the Euler angle representation, the reduced model has a singular point at $\alpha = \pm\pi$.

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \sqrt{1 - q_1{}^2 - q_2{}^2 - q_3{}^2} & q_3 & -q_2 \\ -q_3 & \sqrt{1 - q_1{}^2 - q_2{}^2 - q_3{}^2} & q_1 \\ q_2 & -q_1 & \sqrt{1 - q_1{}^2 - q_2{}^2 - q_3{}^2} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \tag{2.0}$$

$$= \tfrac{1}{2} Q(q_1, q_2, q_3) w$$

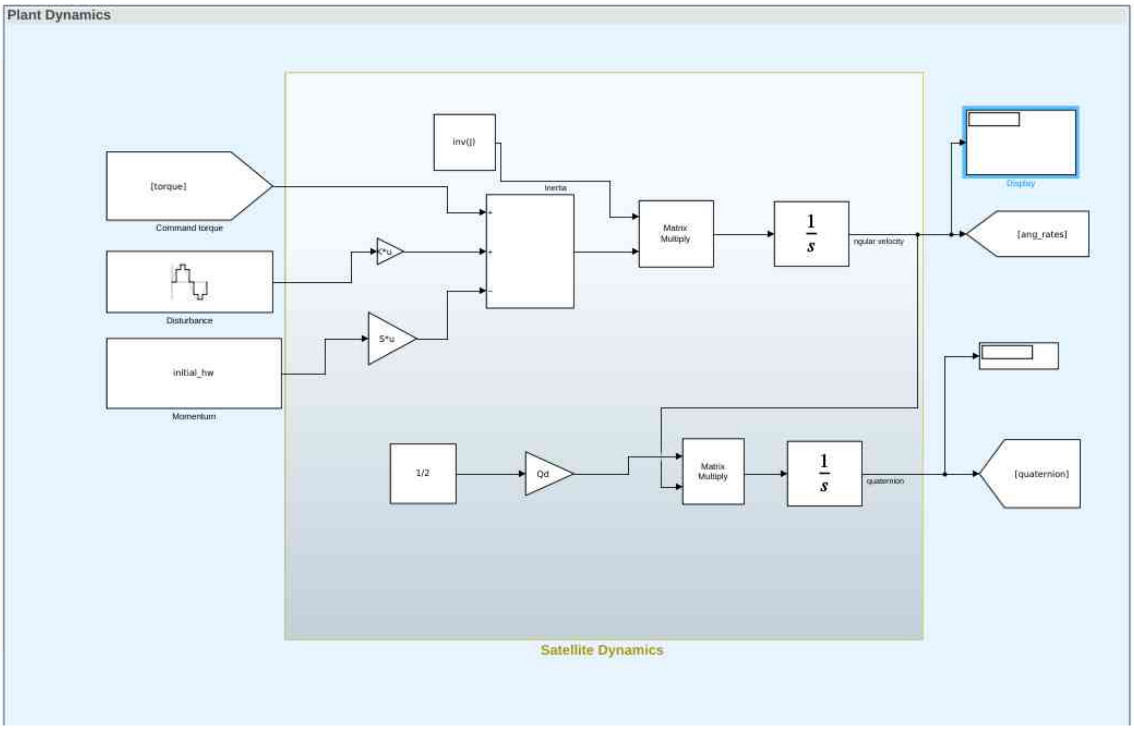

Figure 1 Kinematic and Dynamic model of a spacecraft

The above figure shows the Dynamic and Kinematic model in Simulink. The equation 1.8 shows the kinematic equation of motion using reduced quaternion

$$J * \dot{w} = -w \times (JW) + T + t_d + u$$

The angular velocity vector obtained from dynamic model is included in kinematic model to obtain the updated reduced quaternion

$$\dot{q} = -\frac{1}{2} w^T \times q$$

Optimal (non-PID) quaternion-based control

Using a linearized reduced quaternion model, we can derive an analytical formula for LQR optimal control that is explicitly related to the cost matrices Q and R. The LQR feedback controller globally stabilizes the original nonlinear spacecraft.

The linear system given as

$$\dot{\boldsymbol{x}} = \boldsymbol{Ax} + \boldsymbol{Bu}$$

$$\boldsymbol{y} = \boldsymbol{cx}$$

The feedback matrix for the LQR to minimize the objective function O

$$u = -[R^{-1} J^{-1} F_{11} \quad R^{-1} J^{-1} F_{12}]x = -[D, K]x$$

$$O = \frac{1}{2} \int_0^\infty (x^T Q x + u^T R u) dt$$

Where Q and R are positive definite matrices, $x^T Q x$ cost of deviation from desired equilibrium point. $u^T R u$ cost of energy consumption.

```matlab
%%The simulink model parameters for spacecraft

J=[1.10 0.05 0.00
   0.05 1.9 -0.01
   0.00 -0.01 1.12];                                    %Inertia matrix
Wi=[0.1745 -0.3491 0.2094].';                           %Angular velocity
Wf=[0 0 0].';
quaternion=[0.8812 -0.1687 0.2926];                      %Quaternion initial
qn=sqrt(1-quaternion(1)^2-quaternion(2)^2-quaternion(3)^2);
q_desired=[-0.06680 0.0481 -0.7296 -0.1385].';           %Quaternion final
ang_rates=[0.1745 -0.3491 0.2094].';
initial_hw=J*Wi;
Qd=[-0.4078 -0.2926 -0.1687
    0.2926 -0.4078 -0.8812
    0.1687 0.8812 -0.4078];
S=[0 0.2094 0.3491
   -0.2094 0 0.1745
   -0.3491 -0.1745 0];
td=[1 0 0].';


A=[zeros(3) zeros(3);0.5*eye(3) zeros(3)];
B=[inv(J);zeros(3)];
Q=0.6*eye(6);
R=0.3*eye(3);
K=-lqr(A,B,Q,R);
sim('Satellite_Dynamics_V2')
```

Figure 2 MATLAB Code for LQR

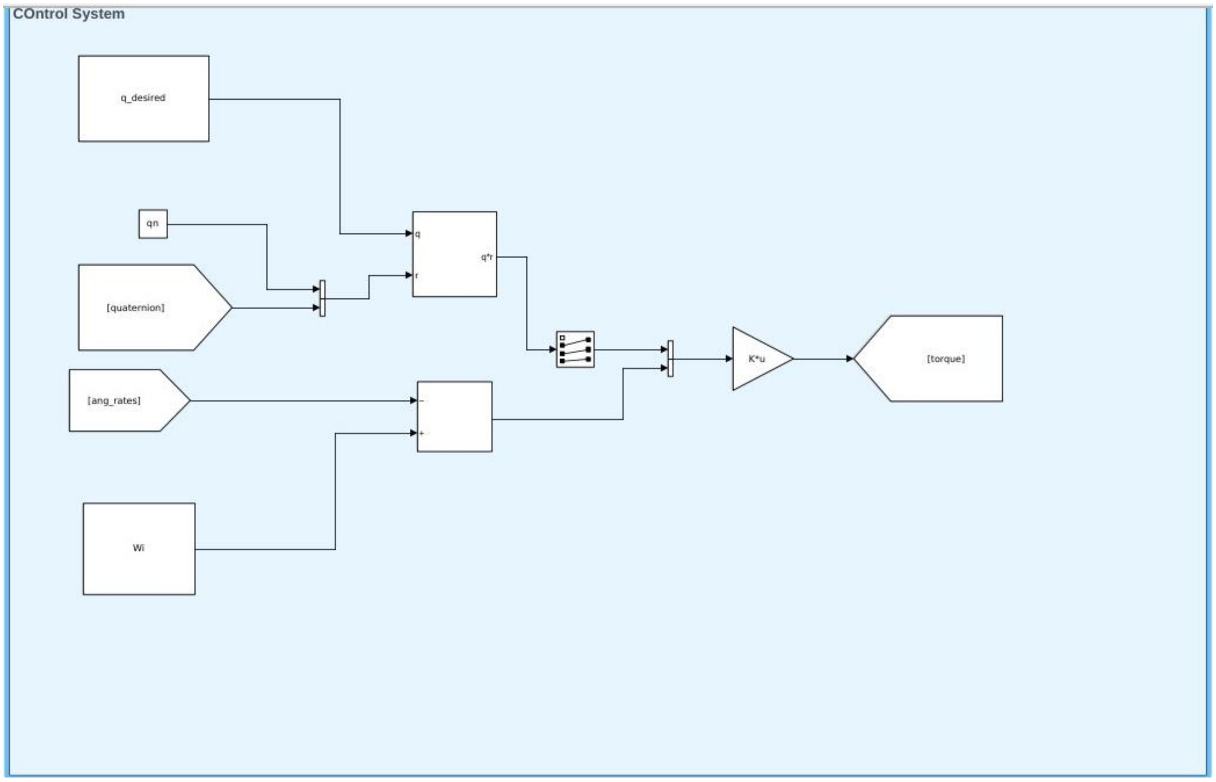

Figure 3 Control system

The above figure represents the control systems of spacecraft.

Quaternion error is obtained from the desired quaternion and the quaternion obtained from the Kinematic model.

$$q_e = q^{-1} * q_n \,(or) q^* * q_n$$
$$w_e = w_i - ang\_rates$$

We know the feedback control u. The output of the feedback is torque

$$u = -k * x$$

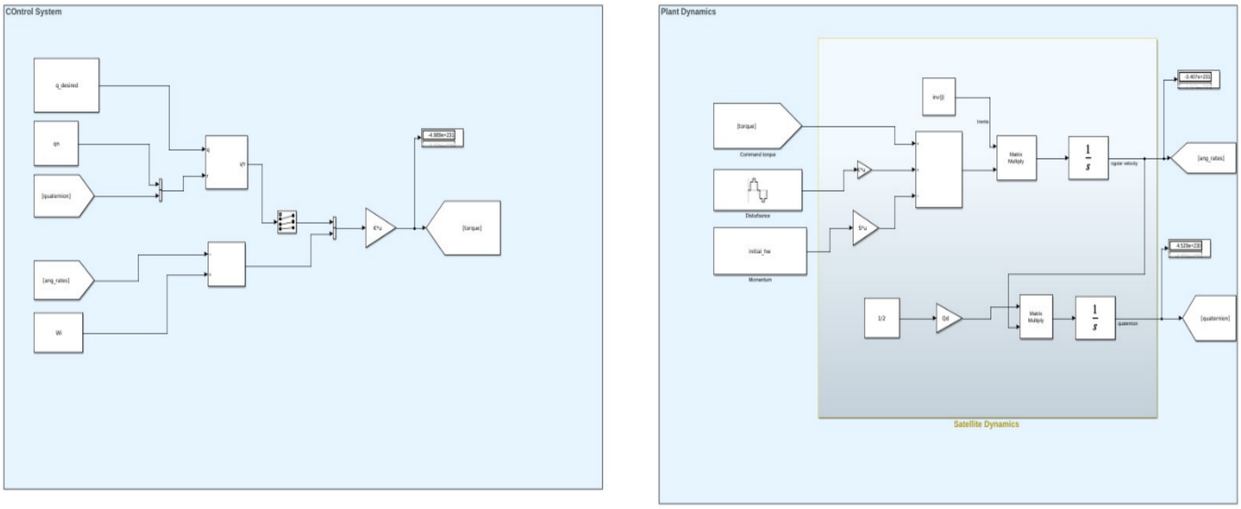

Figure 4 Control System of a Spacecraft

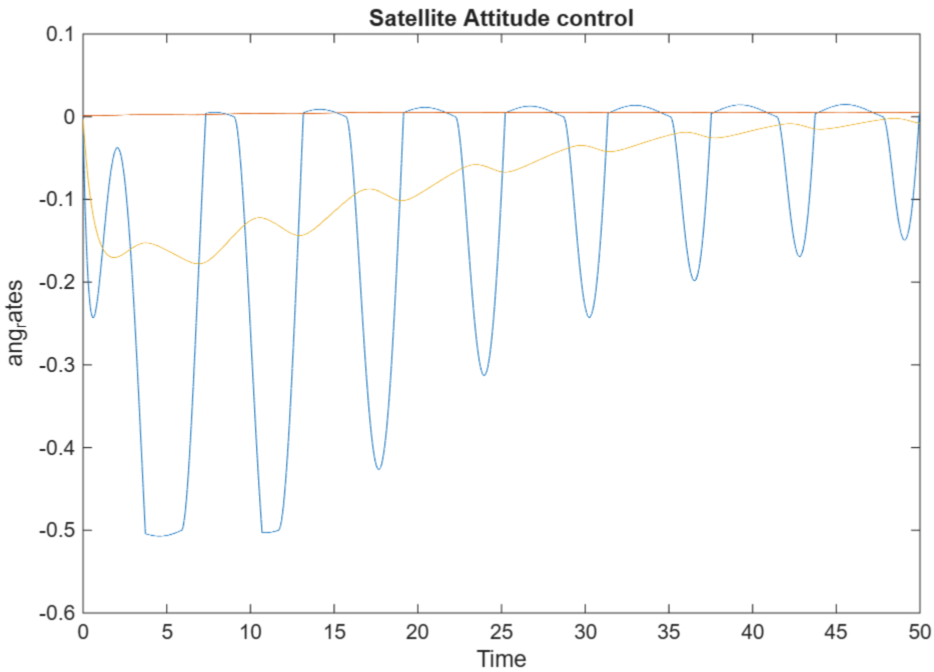

Figure 5 Angular velocity [W1W2W3]

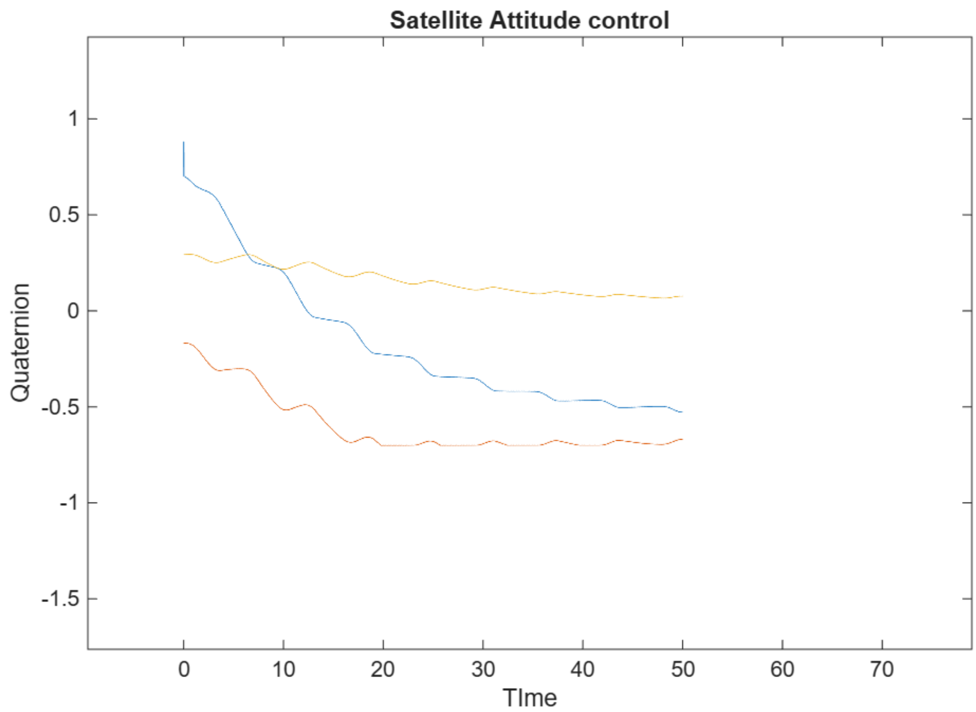

Figure 6Quaternion [q1 q2 q3]

The above graphs fig 5,6 represent convergence to desired angular velocity, and the resulting desired quaternion over a period. This can be further improved by tuning the values Q and R.

Extended Kalman Filter (EKF)

Extended Kalman Filter (EKF) for spacecraft attitude estimation using IMU (gyroscope and accelerometer) and Star Tracker data.

- How does the EKF fuse IMU and Star Tracker measurements to estimate spacecraft attitude?

Inertial Measuring Unit (IMU):

IMUs are used to detect changes in rates of rotation with a high degree of precision. By integrating the rates of rotation over time from a known position, IMUs allow the spacecraft to propagate its attitude when data from other attitude sensors is infrequent.

Star Tracker:

Star trackers determine attitude by taking a picture of the sky and mapping it against a catalogue of known star positions.

IMU Coupling

An attitude estimation system is comprised of IMU for high frequency rate information with periodic star tracker attitude measurement updates. The IMU performance characteristic is represented by the angular random walk (ARW) in degrees per square root hour. It represents how accurate the IMU is and a smaller ARW means less error from IMU measurements. From an initial star tracker measurement, the attitude error will grow quadratically over time as the IMU measurement is propagated.

The star tracker generates cross border sight (CBS) error at each time step. The error for stars estimation of a single star tracker is shown using standard deviation. The combined error of IMU and Star tracker over a multiple time step, a least square weighting is used.

$$\sigma_{tot} = \left(\frac{1}{\sigma_0} + \frac{1}{\sigma_0 + \left(\sigma_{imu}\sqrt{\Delta \mathrm{t}}\right)^2}\right) \tag{3.1}$$

σ_{tot} combined standard deviation
$\sigma_{imu}$is the IMU random walk-in radians per square root second

Extended Kalman Filter (EKF):

Extended Kalman filter is used to linearize a nonlinear system around a prior estimation state. A generic nonlinear system is represented by

$$x_k = f_{k-1}(x_{k-1}, u_{k-1}, \emptyset_{k-1}) \tag{3.2}$$

$$y_k = h_k(x_k, \psi_k) \tag{3.3}$$

x_k is the state vector, u is the input control vector, y is the measurement vector, $\{\emptyset_{k-1}, \Psi_k\}$ are respectively process and measurement noise. The mean noise is assumed to be zero, but the covariance is nonzero as the Kalman filter needs to handle only uncertainty.

$$E[\phi_k] = 0 \quad E[\Psi_k] = 0$$
$$E\left[\phi_k {\phi_j}^T\right] = \delta_{kj} Q_k \qquad E\left[\psi_k \psi_j^T\right] = \delta_{kj} R_k$$

The equations 3.1,3.2 can be linearized using Taylor series expansion after solving. The noises of linearized equations are given by

$$\phi \sim (0, Q_k)$$
$$\psi \sim (0, \mathrm{R}_k)$$

where

$$Q_k = L Q_k L^T \tag{3.4}$$
$$\mathrm{R}_k = M \mathrm{R}_k M^T \tag{3.5}$$

Therefore, we can use the linear Kalman filter equations to estimate the state. This gives the discrete-time extended Kalman filter:

$$\hat{x}_{\frac{k}{k}} = \hat{x}_{\frac{k}{k-1}} + k_k \left[y_k - H_k \hat{x}_{\frac{k}{k-1}} - z_k\right] \tag{3.6}$$

y_k Predicted measurement
$\hat{x}_{\frac{k}{k}}$ Predicted state updated estimate
$\hat{x}_{\frac{k}{k-1}}$ Predicted state previous instant of time
k_k Kalman gain

Extended Kalman Filter state estimation using quaternion:

Quaternion Kinematic model is used for attitude estimation and dynamic model for rotation rates included Kalman filter modelling. The spacecraft model with gaussian noise is considered

$$\dot{w} = -J^{-1} w \times (Jw) + J^{-1} u + \phi_1 \tag{3.7}$$

$$\dot{q} = \frac{1}{2}\Omega(w + \phi_2) \tag{3.8}$$

q is the vector part of the quaternion, ω is the spacecraft rotational rate with respect to the inertial frame, $\phi = [\phi_1 \quad \phi_2]^T$ is the process Gaussian noise, J is the inertia matrix of the spacecraft, and Ω is a matrix given by

$$\Omega = \begin{bmatrix} g(q) & -q_3 & q_2 \\ q_3 & g(q) & -q_1 \\ -q_2 & q_1 & g(q) \end{bmatrix}$$

With $g(q) = \sqrt{1 - {q_1}^2 - {q_2}^2 - {q_3}^2}$

The angular rate measurement w_y and the quaternion measurement q_y. With Star tracker and IMU measurement onboard resulting measurement equation as

$$\dot{\beta} = \emptyset_3$$
$$w_y = w + \beta + \Psi_1$$

For star tracker the bore sight correction angle is applied to the observed quaternion or spacecraft position quaternion to obtain the adjusted quaternion.

$$q_{adj} = q_b \otimes q_{obs}$$

q_b bore sight quaternion
q_{obs} Spacecraft quaternion

Then the quaternion is rotated w.r.t to the error state of star tracker, $\Delta\theta$ is related to truth state and its current estimate as

$$\theta_{true} = \theta_{est} + \Delta\theta$$

The corresponding quaternion error correction

$$dq_b = \begin{bmatrix} \hat{S}\sin\left(\frac{\Delta\theta}{2}\right) \\ \cos\left(\frac{\Delta\theta}{2}\right) \end{bmatrix}$$

Now q_y be the measurement

$$q_y = dq_b \otimes q_{adj}$$
$$q_y = q + \Psi_2$$

The overall system equations are as follows:

$$\dot{w} = -J^{-1}w \times (Jw) + J^{-1}u + \phi_1 \tag{3.7}$$
$$\dot{q} = \frac{1}{2}\Omega(w + \phi_2) \tag{3.8}$$
$$\dot{\beta} = \emptyset_3$$
$$w_y = w + \beta + \Psi_1$$
$$q_y = q + \Psi_2$$

It is rewritten as a standard state space model

$$\dot{x} = f(x, u, \emptyset)$$
$$y = Hx + \Psi$$

this follows similar to the equations from above 3.2,3.3 to estimate the state and process noise further

$$F_{k-1} = \begin{bmatrix} I - J^{-1}(w \times J - (Jw))dt & 0_3 & 0_3 \\ \frac{\partial F_2}{\partial w} & \frac{\partial F_2}{\partial q} & 0_3 \\ 0_3 & 0_3 & I_3 \end{bmatrix}$$

$$L_{k-1} = \begin{bmatrix} I_3 & 0_3 & 0_3 \\ 0_3 & \frac{1}{2}\Omega_{k-1} & 0_3 \\ 0_3 & 0_3 & I_3 \end{bmatrix}$$

How do estimation errors impact the performance of the control law?

The estimation errors have a direct impact on Kalman filter gain K_k

- If the noise is higher the Kalman gain is reduced so that the estimation depends more on system dynamics.
- If the noise is lower the Kalman gain increases so that the estimation depends more on the measurement.